

## EXPERIMENT-2

Implement the Candidate Elimination algorithm to find the Version Space (Specific and General Hypotheses).

### Code:

```
import pandas as pd
import numpy as np
from google.colab import files
print("="*70)
print("    CANDIDATE ELIMINATION ALGORITHM")
print("="*70)
uploaded = files.upload()
data = pd.read_csv("EnjoySport.csv")
print("\nTraining Dataset:\n")
print(data)
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
specific = concepts[0].copy()
general = [["?" for i in range(len(specific))] for j in range(len(specific))]
print("\nInitial Specific Hypothesis:")
print(specific)
print("\nInitial General Hypothesis:")
for g in general:
    print(g)
for i, h in enumerate(concepts):
    print("\n" + "-"*60)
    print("Training Example", i+1)
    print(h)
    if target[i] == "Yes":
```

```

print("Positive Example")
for x in range(len(specific)):
    if h[x] != specific[x]:
        specific[x] = '?'
        general[x][x] = '?'
else:
    print("Negative Example")
    for x in range(len(specific)):
        if h[x] != specific[x]:
            general[x][x] = specific[x]
        else:
            general[x][x] = '?'
print("\nSpecific Hypothesis:")
print(specific)
print("\nGeneral Hypothesis:")
for g in general:
    print(g)
general = [g for g in general if g != ['?']*len(specific)]
print("\n"+"="*70)
print("FINAL SPECIFIC HYPOTHESIS")
print(specific)
print("\nFINAL GENERAL HYPOTHESIS")
for g in general:
    print(g)

```

### Sample Input:

```

... =====
                CANDIDATE ELIMINATION ALGORITHM
=====
Choose Files EnjoySport.csv
EnjoySport.csv(text/csv) - 446 bytes, last modified: 8/5/2026 - 100% done
Saving EnjoySport.csv to EnjoySport (1).csv

```

### Sample Output:

Training Dataset:

...

|   | Sky      | AirTemp | Humidity | Wind   | Water | Forecast | EnjoySport |
|---|----------|---------|----------|--------|-------|----------|------------|
| 0 | Sunny    | Warm    | Normal   | Strong | Warm  | Same     | Yes        |
| 1 | Sunny    | Warm    | High     | Strong | Warm  | Same     | Yes        |
| 2 | Rainy    | Cold    | High     | Strong | Warm  | Change   | No         |
| 3 | Sunny    | Warm    | High     | Strong | Cool  | Change   | Yes        |
| 4 | Sunny    | Cold    | Normal   | Weak   | Warm  | Same     | Yes        |
| 5 | Rainy    | Warm    | Normal   | Weak   | Cool  | Change   | No         |
| 6 | Overcast | Warm    | High     | Strong | Warm  | Same     | Yes        |
| 7 | Overcast | Cool    | Normal   | Weak   | Cool  | Same     | Yes        |
| 8 | Rainy    | Cool    | High     | Strong | Cool  | Change   | No         |
| 9 | Sunny    | Warm    | Normal   | Weak   | Warm  | Same     | Yes        |

Initial Specific Hypothesis:

```
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
```

Initial General Hypothesis:

```
[ '?', '?', '?', '?', '?', '?' ]
[ '?', '?', '?', '?', '?', '?' ]
[ '?', '?', '?', '?', '?', '?' ]
[ '?', '?', '?', '?', '?', '?' ]
[ '?', '?', '?', '?', '?', '?' ]
[ '?', '?', '?', '?', '?', '?' ]
```

### Training Example 1

```
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
```

### Positive Example

Specific Hypothesis:

```
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
```

General Hypothesis:

```
[ '?', '?', '?', '?', '?', '?']
[ '?', '?', '?', '?', '?', '?']
[ '?', '?', '?', '?', '?', '?']
[ '?', '?', '?', '?', '?', '?']
[ '?', '?', '?', '?', '?', '?']
[ '?', '?', '?', '?', '?', '?']
```

Training Example 2  
['Sunny' 'Warm' 'High' 'Strong' 'Warm' 'Same']  
\*\*\* Positive Example

Specific Hypothesis:  
['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

General Hypothesis:  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']

-----  
Training Example 3  
['Rainy' 'Cold' 'High' 'Strong' 'Warm' 'Change']  
Negative Example

Specific Hypothesis:  
['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

General Hypothesis:  
['Sunny', '?', '?', '?', '?', '?']  
['?', 'Warm', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', 'Same']

-----  
Training Example 4  
['Sunny' 'Warm' 'High' 'Strong' 'Cool' 'Change']  
Positive Example

Specific Hypothesis:  
['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:  
['Sunny', '?', '?', '?', '?', '?']  
['?', 'Warm', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
.....

General Hypothesis:  
... ['Sunny', '?', '?', '?', '?', '?']  
['?', 'Warm', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']

-----  
Training Example 5  
['Sunny' 'Cold' 'Normal' 'Weak' 'Warm' 'Same']  
Positive Example

Specific Hypothesis:  
['Sunny' '?' '?' '?' '?' '?']

General Hypothesis:  
['Sunny', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']

-----  
Training Example 6  
['Rainy' 'Warm' 'Normal' 'Weak' 'Cool' 'Change']  
Negative Example

Specific Hypothesis:  
['Sunny' '?' '?' '?' '?' '?']

General Hypothesis:  
['Sunny', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']

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Training Example 7  
['Overcast' 'Warm' 'High' 'Strong' 'Warm' 'Same']

Specific Hypothesis:

['?' '?' '?' '?' '?' '?']

...

General Hypothesis:

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

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Training Example 8

['Overcast' 'Cool' 'Normal' 'Weak' 'Cool' 'Same']

Positive Example

Specific Hypothesis:

['?' '?' '?' '?' '?' '?']

General Hypothesis:

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

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Training Example 9

['Rainy' 'Cool' 'High' 'Strong' 'Cool' 'Change']

Negative Example

Specific Hypothesis:

['?' '?' '?' '?' '?' '?']

General Hypothesis:

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

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Training Example 10

['Sunny' 'Warm' 'Normal' 'Weak' 'Warm' 'Same']

Positive Example

Specific Hypothesis:

['?' '?' '?' '?' '?' '?']

General Hypothesis:

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

['?', '?', '?', '?', '?', '?']

=====

FINAL SPECIFIC HYPOTHESIS

['?' '?' '?' '?' '?' '?']

FINAL GENERAL HYPOTHESIS